**ITI0402 Loogika, sügis 2026**
**Harjutus 9**
**Lahendused**

1.

$$\dfrac{\dfrac{\dfrac{\overset{+1}{\forall x\ p(x)}}{p(f(x'))}\ \forall\mathcal{E}}{\forall x\ p(f(x))}\ \forall\mathcal{I}, x' \text{ värske}}{\forall x\ p(x) \supset \forall x\ p(f(x))}\ \supset\mathcal{I}, -1$$

$$\dfrac{\dfrac{\dfrac{\dfrac{\overset{+1}{\forall x\ p(x, f(x))}}{p(y', f(y'))}\ \forall\mathcal{E}}{\exists z\ p(y', z)}\ \exists\mathcal{I}}{\forall y\ \exists z\ p(y, z)}\ \forall\mathcal{I}, x' \text{ värske}}{\forall x\ p(x, f(x)) \supset \forall y\ \exists z\ p(y, z)}\ \supset\mathcal{I}, -1$$

$$\dfrac{\dfrac{\dfrac{\overset{+1}{\neg\exists x\ p(x)} \quad \dfrac{\overset{+2}{p(x')}}{\exists x\ p(x)}\ \exists\mathcal{I}}{\bot}\ \neg\mathcal{E}}{\neg p(x')}\ \neg\mathcal{I}, -2}{\forall x\ \neg p(x)}\ \forall\mathcal{I}, x' \text{ värske}}{\neg\exists x\ p(x) \supset \forall x\ \neg p(x)}\ \supset\mathcal{I}, -1$$

$$\dfrac{\dfrac{\overset{+1}{p(a) \vee p(b)} \quad \dfrac{\overset{+2}{p(a)}}{\exists x\ p(x)}\ \exists\mathcal{I} \quad \dfrac{\overset{+3}{p(b)}}{\exists x\ p(x)}\ \exists\mathcal{I}}{\exists x\ p(x)}\ \vee\mathcal{E}, -2, -3}{p(a) \vee p(b) \supset \exists x\ p(x)}\ \supset\mathcal{I}, -1$$

$$\dfrac{\dfrac{\overset{+1}{\exists x\ (p(x) \vee q(x))} \quad \dfrac{\overset{+2}{p(x') \vee q(x')} \quad \dfrac{\dfrac{\overset{+3}{p(x')}}{\exists x\ p(x)}\ \exists\mathcal{I}}{\exists x\ p(x) \vee \exists x\ q(x)}\ \vee\mathcal{I}_L \quad \dfrac{\dfrac{\overset{+r}{q(x')}}{\exists x\ q(x)}\ \exists\mathcal{I}}{\exists x\ p(x) \vee \exists x\ q(x)}\ \vee\mathcal{I}_R}{\exists x\ p(x) \vee \exists x\ q(x)}\ \vee\mathcal{E}, -3, -4}{\exists x\ p(x) \vee \exists x\ q(x)}\ \exists\mathcal{E}, -2, x' \text{ värske}}{\exists x\ (p(x) \vee q(x)) \supset \exists x\ p(x) \vee \exists x\ q(x)}\ \supset\mathcal{I}, -1$$

$$\dfrac{\dfrac{\overset{+1}{\exists x\ p(x)}}{\exists x\ p(x) \vee \forall x\ \neg p(x)}\ \vee\mathcal{I}_L \quad \dfrac{\dfrac{\dfrac{\dfrac{\overset{+2}{\neg\exists x\ p(x)} \quad \dfrac{\overset{+3}{p(x')}}{\exists x\ p(x)}\ \exists\mathcal{I}}{\bot}\ \neg\mathcal{E}}{\neg p(x')}\ \neg\mathcal{I}, -3}{\forall x\ \neg p(x)}\ \forall\mathcal{I}, x' \text{ värske}}{\exists x\ p(x) \vee \forall x\ \neg p(x)}\ \vee_R}{\exists x\ p(x) \vee \forall x\ \neg p(x)}\ \text{Dil.}, -1, -2$$

$$\dfrac{\dfrac{\dfrac{}{\exists x\ \neg p(x)}\,{\scriptstyle +1}}{\exists x\ \neg p(x) \vee \forall x\ p(x)}\ \vee\mathcal{I}_L \qquad \dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{p(x')}\,{\scriptstyle +3} \qquad \dfrac{\dfrac{}{\neg p(x')}\,{\scriptstyle +4}}{\exists x\ \neg p(x)}\ \exists\mathcal{I}}{\bot}\ \neg\mathcal{E}}{p(x')}\ \bot\mathcal{E}}{\forall x\ p(x)}\ \text{Dil.}, -3, -4}{\exists x\ \neg p(x) \vee \forall x\ p(x)}\ \forall\mathcal{I}, x' \text{ värske}}{\exists x\ \neg p(x) \vee \forall x\ p(x)}\ \vee_R \qquad \dfrac{\dfrac{\dfrac{}{\neg\exists x\ \neg p(x)}\,{\scriptstyle +2} \qquad \dfrac{\dfrac{}{\neg p(x'')}\,{\scriptstyle +6}}{\exists x\ \neg p(x)}\ \exists\mathcal{I}}{\bot}\ \neg\mathcal{E}}{p(x'')}\ \bot\mathcal{E}}{\forall x\ p(x)}\ \text{Dil.}, -5, -6}{\exists x\ \neg p(x) \vee \forall x\ p(x)}\ \forall\mathcal{I}, x'' \text{ värske}}{\exists x\ \neg p(x) \vee \forall x\ p(x)}\ \text{Dil.}, -1, -2$$

$$\dfrac{\dfrac{\dfrac{}{\exists x\ \neg p(x)}\,{\scriptstyle +1} \qquad \dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\neg p(x')}\,{\scriptstyle -1} \qquad \dfrac{}{p(x')}\,{\scriptstyle -3}}{\bot}\ \neg\mathcal{E}}{\forall x\ p(x)}\ \bot\mathcal{E}}{p(x') \supset \forall x\ p(x)}\ \supset\mathcal{I}, -3}{\exists x\ (p(x) \supset \forall x\ p(x))}\ \exists\mathcal{I}}{\exists x\ (p(x) \supset \forall x\ p(x))}\ \exists\mathcal{E}, x' \text{ värske} \qquad \dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{p(x'')}\,{\scriptstyle +5}}{\forall x\ p(x)}\ \forall\mathcal{I}, x'' \text{ värske}}{p(x') \supset \forall x\ p(x)}\ \supset\mathcal{I}, -4}{\exists x\ (p(x) \supset \forall x\ p(x))}\ \exists\mathcal{I}}{\exists x\ (p(x) \supset \forall x\ p(x))}\ \text{Dil.}, -1, -2$$